Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LEARNER GUIDE

For

Google Professional Machine Learning Engineer Training

Course Code: C740

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 8.0

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
7.0	Legacy release	Original course material and 89-slide source deck.	Tertiary Infotech Academy
8.0	12 September 2026	Current six-domain Google PMLE alignment; 415 visual slides; 16 detailed Activities; account setup, Vertex AI, TensorFlow, serving, MLOps and monitoring expansion.	Dr Alfred Ang

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

TABLE OF CONTENTS

Course Learning Outcomes **6**

Skills Framework Alignment **6**

Knowledge statements 6

Ability statements 6

How to Use This Guide **6**

Current Google PMLE Exam Domains **7**

Technical Reference Chapters **7**

Domain 1 — Architecting low-code AI solutions (~13%) **7**

BigQuery ML model selection 8

AutoML tabular workflow 9

Google Cloud AI APIs 11

Model Garden and foundation model selection 12

Gemini tuning with BigQuery 14

Prompt and context engineering 15

Responsible low-code design 17

Cost, latency and availability architecture 18

Domain 2 — Collaborating to manage data and models (~16%) **20**

Data organization by modality 20

Preprocessing tool selection 22

Feature engineering and selection 23

Feature Store architecture 25

Privacy and sensitive-data controls 26

Secure Workbench and Colab Enterprise 28

Experiment tracking and lineage 29

Predictive and gen AI evaluation 31

Domain 3 — Scaling prototypes into ML models (~21%) **32**

Model type and product decision 32

Training data ingestion 34

TensorFlow and Keras training 35

Vertex AI custom training 37

Hyperparameter tuning 38

Foundation model fine-tuning 40

CPU, GPU and TPU selection 41

Distributed training 43

Domain 4 — Serving and scaling models (~20%) **44**

Batch versus online inference 44

Prebuilt and custom serving containers 46

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Model Registry and versioning....47
Canary and A/B rollout....49
Inference preprocessing and postprocessing....50
Public and private endpoints....52
Autoscaling and serving hardware....53
Cloud Run and GKE serving....55
Domain 5 — Automating and orchestrating ML pipelines (~18%)....56
Pipeline component contracts....56
Vertex AI Pipelines and Kubeflow....58
Cloud Composer and Airflow....60
Ray on Vertex AI....61
Training-serving preprocessing parity....63
Retraining policy....64
CI/CD/CT with Cloud Build....66
Pipeline governance and reproducibility....67
Domain 6 — Monitoring AI solutions (~13%)....69
AI security and Model Armor....69
Responsible AI and bias monitoring....70
Explainability and feature attribution....72
Data drift, skew and concept drift....73
Gen AI evaluation in production....75
SLOs, logs and alerts....76
Incident troubleshooting....78
Cost and performance optimization....79
Detailed Activity Guides....81
Activity 01 — Create a Safe Google Cloud Training Environment....81
Activity 02 — Draft an ML Architecture Decision Record....87
Activity 03 — Train and Evaluate a BigQuery ML Churn Model....92
Activity 04 — Design an AutoML Tabular Experiment....97
Activity 05 — Compare Pretrained Google Cloud AI APIs....102
Activity 06 — Build a Data Quality and Feature Pipeline....107
Activity 07 — Train a TensorFlow/Keras Model....112
Activity 08 — Package a Vertex AI Custom Training Job....116
Activity 09 — Analyze Vertex AI Hyperparameter Trials....121
Activity 10 — Plan Distributed PyTorch GPU Training....125
Activity 11 — Verify Feature Store Point-in-Time Parity....130
Activity 12 — Register and Canary a Model Version....134
Activity 13 — Serve a Model on Cloud Run....139

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activity 14 — Compile a Vertex AI Pipeline....144
Activity 15 — Design an ML CI/CD/CT Release....148
Activity 16 — Investigate Drift, Bias and an ML Incident....153
Source Register....158

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Course Learning Outcomes

- LO1: Assess and draft Google Cloud solution specifications to meet expected business machine learning requirements.
- LO2: Develop implementation plans and resolve issues for Google Cloud machine learning solutions.
- LO3: Develop and review processes for metrics associated with Google Cloud machine learning solution implementation.

How to Use This Guide

1. Read the relevant domain mechanism before its Activity.
2. Work from the Activity folder so relative paths are predictable.
3. Use mock data first. Execute paid Google Cloud resources only with trainer approval.
4. Preserve evidence: configuration, job/revision ID, metrics, decision and cleanup result.
5. Never place credentials, service-account keys, personal data or production prompts in the evidence pack.

Current Google PMLE Exam Domains

The course follows the official Google Cloud Professional Machine Learning Engineer exam guide current on the release date. Approximate weights total about 101% because Google publishes rounded values.

- Domain 1 — Architecting low-code AI solutions (~13%). Legacy coverage: Topics 1–3: GCP fundamentals, Google ML services, ML workflow.
- Domain 2 — Collaborating to manage data and models (~16%). Legacy coverage: Topics 1, 3 and 5: big data, data quality, feature engineering.
- Domain 3 — Scaling prototypes into ML models (~21%). Legacy coverage: Topics 4–6: TensorFlow, feature engineering, enterprise ML.
- Domain 4 — Serving and scaling models (~20%). Legacy coverage: Topics 6–7: batch/online prediction, production systems.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Domain 5 — Automating and orchestrating ML pipelines (~18%). Legacy coverage: Topic 8: MLOps, CI/CD and repeatable workflows.
- Domain 6 — Monitoring AI solutions (~13%). Legacy coverage: Topics 6–8: testing, monitoring, troubleshooting and MLOps.

Domain 1 — Architecting low-code AI solutions (~13%)

Topics 1–3: GCP fundamentals, Google ML services, ML workflow.

BigQuery ML model selection

Mechanism flow

Business question → BigQuery table → CREATE MODEL → ML.EVALUATE → ML.PREDICT

Service or method decision

- Logistic regression — binary outcome with calibrated probability
- Boosted trees — non-linear tabular relationships
- ARIMA_PLUS — forecasting with seasonality

Runnable configuration pattern

```
CREATE OR REPLACE MODEL `ml.churn`
OPTIONS(model_type='LOGISTIC_REG', input_label_cols=['churned']) AS
SELECT * EXCEPT(customer_id) FROM `ml.customer_features`;
```

Metric: Precision–recall trade-off

Formula/decision rule: precision = TP/(TP+FP); recall = TP/(TP+FN)

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
High accuracy, poor recall	class imbalance	use PR AUC and class weights
Query cost spike	unbounded scans	partition, cluster and select columns
Leakage	post-outcome feature	point-in-time feature review

Evidence chain

- SQL text and job ID
- ML.EVALUATE result
- confusion matrix
- model metadata and labels

Exam-style decision case

A retailer needs churn scores from 40 TB already in BigQuery with no custom serving code.

Best decision: Use BigQuery ML; keep training and scoring beside governed data and batch predictions.

Related practical: Activity 03

AutoML tabular workflow

Mechanism flow

Schema → Target → Train split → Budget → Evaluate → Register

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Service or method decision

- AutoML — strong baseline and limited ML engineering capacity
- Custom training — custom loss or architecture
- BigQuery ML — SQL team and warehouse-resident data

Runnable configuration pattern

```
gcloud ai models upload --region=$REGION --display-name=churn-automl \
+  --container-image-uri=$PREDICTION_IMAGE --artifact-uri=$MODEL_DIR
```

Metric: Training budget curve

Formula/decision rule: marginal gain = metric(t)-metric(t-1)

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Invalid target	nulls or unsupported label	profile and clean label
Weak validation	random split on temporal data	use time-aware split
Over-budget	no stopping criterion	cap node hours and monitor marginal gain

Evidence chain

- dataset schema
- split manifest
- training budget
- evaluation slices
- registered model version

Exam-style decision case

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

A lending team needs a governed risk baseline quickly, but predictions must be explainable by segment.

Best decision: Start with AutoML tabular, require slice metrics and feature attributions before deployment.

Related practical: Activity 04

Google Cloud AI APIs

Mechanism flow

Content → Auth → API request → Structured response → Human review

Service or method decision

- Document AI — forms, invoices and OCR structure
- Vision API — labels, OCR and image properties
- Translate API — managed multilingual translation

Runnable configuration pattern

```
curl -s -X POST -H "Authorization: Bearer $(gcloud auth print-access-token)" \
+  -H 'Content-Type: application/json' \
+  "https://vision.googleapis.com/v1/images:annotate" -d @request.json
```

Metric: API confidence threshold

Formula/decision rule: review_rate = count(score < threshold)/N

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
403	API disabled or IAM missing	enable API; grant narrow role
429	quota exhausted	backoff, batch, request quota
Low confidence	domain mismatch	human review or custom model

Evidence chain

- request JSON
- HTTP status
- operation ID
- confidence distribution
- review log

Exam-style decision case

An insurer must extract line items from varied claim forms and retain reviewer evidence.

Best decision: Use Document AI processor plus confidence-based human review; do not build OCR from scratch.

Related practical: Activity 05

Model Garden and foundation model selection

Mechanism flow

Task → Constraints → Candidate models → Evaluation set → Endpoint

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Service or method decision

- Gemini managed model — multimodal and managed operations
- Open model — weight control or special license needs
- Specialized API — narrow commodity capability

Runnable configuration pattern

```
MODEL_ID='gemini-2.5-flash'
P95_MS=850
MAX_COST_PER_1K=0.020
SAFETY='BLOCK_MEDIUM_AND_ABOVE'
```

Metric: Utility score

Formula/decision rule: 0.45*quality - 0.25*cost - 0.20*latency + 0.10*availability

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Benchmark mismatch	public task differs	use production-like eval set
License risk	terms not reviewed	legal and model-card review
Region gap	model unavailable	confirm region before design

Evidence chain

- candidate matrix
- model card
- evaluation dataset
- quality/cost/latency report
- approval record

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Exam-style decision case

A support assistant must ground answers in policy PDFs with sub-second median response.

Best decision: Evaluate managed Gemini candidates on groundedness, latency and cost; select by measured SLOs.

Related practical: Activity 02

Gemini tuning with BigQuery

Mechanism flow

Curate examples → Split → Tune → Evaluate → Publish

Service or method decision

- Prompt only — behavior adjustable with instructions
- Supervised tuning — stable style/task examples
- RAG — facts change and citations matter

Runnable configuration pattern

```
SELECT input_text, output_text
FROM `genai.tuning_examples`
WHERE approved = TRUE AND split IN ('TRAIN','VALIDATION');
```

Metric: Tuning generalization gap

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Formula/decision rule: gap = train_score - validation_score

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Memorization	duplicates or tiny set	deduplicate and widen validation
Unsafe output	bad examples	curation and safety eval
No gain	task is knowledge-bound	use RAG instead

Evidence chain

- approved tuning table
- data lineage
- base/tuned comparison
- safety results
- versioned endpoint

Exam-style decision case

A legal drafting model must follow a house style, while statute facts remain retrieval-grounded.

Best decision: Tune for stable style and use RAG for changing legal facts; evaluate each separately.

Related practical: Activity 04

Prompt and context engineering

Mechanism flow

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

System instruction → Retrieved context → User task → Tool call → Grounded answer

Service or method decision

- Few-shot — stable pattern with representative examples
- RAG — current private knowledge
- Tool use — actions or authoritative calculations

Runnable configuration pattern

```
SYSTEM='Use only supplied context; cite source_id.'
CONTEXT_TOP_K=5
TEMPERATURE=0.1
MAX_OUTPUT_TOKENS=800
```

Metric: Groundedness by retrieval depth

Formula/decision rule: grounded = supported_claims / total_claims

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Hallucination	context missing or ignored	abstain and require citations
Prompt injection	untrusted retrieved text	isolate instructions from data
Context bloat	too many chunks	rerank and cap tokens

Evidence chain

- prompt template
- retrieval trace
- tool-call log

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- citation coverage
- red-team cases

Exam-style decision case

An internal HR assistant must answer only from approved policies and refuse missing-policy questions.

Best decision: Use RAG with instruction/data separation, citation checks, and a tested abstention path.

Related practical: Activity 02

Responsible low-code design

Mechanism flow

Stakeholders → Harms → Slices → Controls → Monitoring → Appeal

Service or method decision

- Human-in-loop — high-impact or uncertain decisions
- Human-on-loop — automated low-risk with review
- Automated — reversible, validated low-harm tasks

Runnable configuration pattern

```
slices = ['region','age_band','channel']
max_tpr_gap = 0.05
min_group_n = 100
```


© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

```
require_appeal = True
```

Metric: Equal opportunity gap

Formula/decision rule: max(TPR_group)-min(TPR_group)

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Hidden disparity	aggregate-only metric	slice evaluation
Automation bias	no override	human authority and appeal
Purpose creep	reused prediction	use limitation and review

Evidence chain

- impact assessment
- slice metrics
- data card
- human-review SOP
- appeal and incident log

Exam-style decision case

A model prioritizes public-service cases where false negatives have unequal community impact.

Best decision: Require human review, group metrics, documented thresholds and a usable appeal process.

Related practical: Activity 16

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Cost, latency and availability architecture

Mechanism flow

Demand → SLO → Capacity model → Region → Caching → Fallback

Service or method decision

- Batch — large volume without immediate response
- Online endpoint — predictable low latency
- Serverless — bursty stateless inference

Runnable configuration pattern

```
target_p95_ms: 300
min_replicas: 2
max_replicas: 20
cache_ttl_s: 900
regional_failover: true
```

Metric: Capacity headroom

Formula/decision rule: headroom = 1 - peak_qps/max_qps

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Tail latency	cold starts or saturation	min replicas and load test
High cost	oversized accelerator	profile and right-size
Regional outage	single region	tested fallback design

Evidence chain

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- SLO
- load-test curve
- cost estimate
- quota check
- failover test evidence

Exam-style decision case

A recommendation API sees 10x campaign bursts and must sustain 99.9% monthly availability.

Best decision: Use autoscaled online serving with minimum warm capacity, cached features and tested regional fallback.

Related practical: Activity 02

Domain 2 — Collaborating to manage data and models (~16%)

Topics 1, 3 and 5: big data, data quality, feature engineering.

Data organization by modality

Mechanism flow

Sources → Landing zone → Curated tables → Feature views → Training snapshots

Service or method decision

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- BigQuery — structured analytics and SQL
- Cloud Storage — large unstructured objects
- BigLake — governed access across lakes and warehouses

Runnable configuration pattern

```
gsutil -m cp data/images/*.jpg gs://$BUCKET/raw/images/
bq load --autodetect --source_format=CSV ml.customers gs://$BUCKET/raw/customers.csv
```

Metric: Freshness lag

Formula/decision rule: lag = ingestion_time - event_time

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Schema drift	producer changed fields	contract checks
Duplicate entities	weak keys	deduplicate by event id
Training skew	mutable source	snapshot and version

Evidence chain

- bucket layout
- table schema
- partition stats
- snapshot timestamp
- lineage graph

Exam-style decision case

A multimodal claims model uses PDFs, images and tabular policy history from three teams.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Best decision: Land objects in Cloud Storage, curated entities in BigQuery, and bind a point-in-time training manifest.

Related practical: Activity 06

Preprocessing tool selection

Mechanism flow

Volume → Transform → Latency → State → Runner → Output

Service or method decision

- pandas/sklearn — fits memory and iteration speed matters
- Dataflow — managed Beam batch/stream pipelines
- Spark — existing distributed Spark ecosystem

Runnable configuration pattern

```
with beam.Pipeline(options=opts) as p:
  rows = p | beam.io.ReadFromText(uri)
  clean = rows | beam.Map(validate_and_transform)
  clean | beam.io.WriteToBigQuery(table)
```

Metric: Throughput vs workers

Formula/decision rule: records_per_s = processed / elapsed_s

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
OOM	in-memory transform too	distributed runner

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

	large	
Late data	watermark misconfigured	event-time windows
Hot key	skewed grouping	fan-out or rekey

Evidence chain

- data profile
- runner graph
- worker metrics
- dead-letter table
- reconciliation count

Exam-style decision case

A 2 TB daily feed joins clickstream windows and must finish before a 06:00 training job.

Best decision: Use Dataflow batch with autoscaling, dead-letter handling and a measured completion SLO.

Related practical: Activity 06

Feature engineering and selection

Mechanism flow

Raw columns → Transforms → Candidate features → Selection → Validated matrix

Service or method decision

- Standardization — linear and distance models

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Vocabulary/embedding — high-cardinality text
- Cross feature — known interaction with bounded cardinality

Runnable configuration pattern

```
preprocess = ColumnTransformer([
 ('num', StandardScaler(), numeric),
 ('cat', OneHotEncoder(handle_unknown='ignore'), categorical)])
```

Metric: Validation AUC by feature set

Formula/decision rule: lift = AUC_candidate - AUC_baseline

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Leakage	future value encoded	time-aware review
Cardinality blow-up	unbounded categories	hash or vocabulary cap
Train/serve mismatch	duplicate code	shared transform artifact

Evidence chain

- feature definitions
- fit statistics
- selection report
- schema signature
- unit tests

Exam-style decision case

A fraud model sees merchant IDs never observed during training and drifts by region.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Best decision: Use unknown-safe categorical encoding, region slices and versioned preprocessing artifacts.

Related practical: Activity 06

Feature Store architecture

Mechanism flow

Batch sources → Feature pipeline → Offline store → Online store → Entity lookup → Model

Service or method decision

- Offline features — training and batch scoring
- Online features — low-latency current values
- Embedding retrieval — nearest-neighbor similarity

Runnable configuration pattern

```
entity_id: customer_id
feature_view: customer_realtime
features: [txn_count_1h, spend_7d, risk_score]
online_ttl: 3600
```

Metric: Online feature freshness

Formula/decision rule: freshness = now - feature_timestamp

Failure diagnosis

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Observable symptom	Likely mechanism	Control or verification
Stale feature	pipeline lag	freshness alert
Wrong entity	key mismatch	contract and lookup test
Point-in-time leak	latest-value join	as-of join

Evidence chain

- feature registry
- entity schema
- materialization job
- freshness dashboard
- point-in-time test

Exam-style decision case

A real-time fraud endpoint needs 50 ms feature lookups and reproducible historical training values.

Best decision: Use online and offline feature paths with one definition and explicit point-in-time correctness tests.

Related practical: Activity 11

Privacy and sensitive-data controls

Mechanism flow

Discover → Classify → Minimize → Protect → Authorize → Audit

Service or method decision

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- DLP inspection — discover PII at scale
- Tokenization — retain joins without clear identifiers
- Aggregation — reduce re-identification risk

Runnable configuration pattern

```
gcloud dlp jobs create inspect pii-scan --location=global \
+  --storage-config-file=storage.json --inspect-config-file=inspect.json
```

Metric: PII exposure rate

Formula/decision rule: exposure = unprotected_sensitive_fields / sensitive_fields

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Open bucket	inherited broad role	least privilege and policy scan
PII in notebook	local export	managed storage and VPC controls
Log leakage	payload logging	redact before logging

Evidence chain

- DLP findings
- classification tags
- IAM policy
- encryption key record
- audit log query

Exam-style decision case

A healthcare team shares a training dataset across projects while patient identifiers must not leave a boundary.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Best decision: De-identify, tokenize approved join keys, restrict perimeters and verify access in audit logs.

Related practical: Activity 01

Secure Workbench and Colab Enterprise

Mechanism flow

Identity → Network → Runtime → Repository → Artifact store → Shutdown

Service or method decision

- Workbench managed notebook — integrated Vertex AI and controlled project
- Colab Enterprise — managed collaborative notebooks
- Local notebook — only for non-sensitive lightweight work

Runnable configuration pattern

```
disable_public_ip: true
service_account: ml-dev@$PROJECT.iam.gserviceaccount.com
idle_shutdown_minutes: 60
repository: approved-ml-repo
```

Metric: Idle compute waste

Formula/decision rule: waste = idle_instance_hours / total_instance_hours

Failure diagnosis

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Observable symptom	Likely mechanism	Control or verification
Credential sprawl	user keys in cells	service account and ADC
Data exfiltration	public egress	private IP and perimeter
Cost leak	notebook left on	idle shutdown

Evidence chain

- runtime configuration
- IAM binding
- network route
- repository commit
- shutdown event

Exam-style decision case

A regulated analytics team needs collaborative GPUs without public IPs or downloaded service-account keys.

Best decision: Use managed notebooks with private networking, attached least-privilege identity and idle shutdown.

Related practical: Activity 01

Experiment tracking and lineage

Mechanism flow

Run config → Dataset version → Training job → Metrics → Artifact → Registry

Service or method decision

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Vertex AI Experiments — managed run comparison
- ML Metadata — pipeline lineage
- Git + manifest — code and configuration provenance

Runnable configuration pattern

```
run.log_params({'lr':0.001,'batch_size':128,'data_uri':DATA_URI})
run.log_metrics({'val_auc':0.842,'val_loss':0.391})
```

Metric: Validation AUC by run

Formula/decision rule: best = argmax(metric subject to fairness and latency gates)

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Unrepeatable run	missing dataset version	immutable URI and manifest
Metric cherry-pick	no primary metric	pre-register gates
Artifact orphan	manual upload	pipeline metadata

Evidence chain

- commit SHA
- dataset URI
- parameter set
- metrics
- model artifact checksum

Exam-style decision case

Two teams report different best models because they trained on mutable tables and recorded only accuracy.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Best decision: Bind runs to immutable data and code, predefine metrics, and compare through one experiment tracker.

Related practical: Activity 09

Predictive and gen AI evaluation

Mechanism flow

Task rubric → Gold set → Automated metrics → Slice review → Human review → Release gate

Service or method decision

- Classification metrics — labeled predictive outcomes
- LLM-as-judge — scaled rubric scoring with calibration
- Human evaluation — high-stakes nuance and preference

Runnable configuration pattern

```
rubric = {'groundedness':0.35,'correctness':0.35,'safety':0.20,'style':0.10}
release_if = 'score>=0.82 and safety_violations==0'
```

Metric: Evaluation agreement

Formula/decision rule: agreement = matching_ratings / reviewed_items

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Judge bias	uncalibrated model judge	human calibration sample

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Contamination	eval in tuning set	holdout governance
Aggregate masking	no slices	report by task and risk

Evidence chain

- rubric version
- gold-set manifest
- metric results
- reviewer adjudication
- release decision

Exam-style decision case

A gen-AI summarizer improves ROUGE but starts omitting critical contraindications in one document type.

Best decision: Use risk-weighted rubric, document-type slices and human adjudication; block release on critical omissions.

Related practical: Activity 16

Domain 3 — Scaling prototypes into ML models (~21%)

Topics 4–6: TensorFlow, feature engineering, enterprise ML.

Model type and product decision

Mechanism flow

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Task → Data → Constraints → Baseline → Candidate → Decision

Service or method decision

- ARIMA_PLUS — interpretable time-series forecast
- DNN — large nonlinear supervised task
- LLM — language generation or reasoning

Runnable configuration pattern

```
decision = {'task':'forecast','horizon_days':30,'need_intervals':True,
            'product':'BigQuery ML ARIMA_PLUS'}
```

Metric: Decision utility

Formula/decision rule: utility = quality - cost_penalty - latency_penalty - risk_penalty

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Over-complex model	baseline omitted	establish simple baseline
Wrong metric	business loss ignored	map metric to decision
Unclear ownership	product mismatch	operating-model review

Evidence chain

- requirements matrix
- baseline result
- candidate benchmark
- risk review
- architecture decision record

Exam-style decision case

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

A planner needs 30-day demand intervals from structured history and must explain seasonality.

Best decision: Choose ARIMA_PLUS in BigQuery ML before a DNN; validate interval coverage and holiday effects.

Related practical: Activity 02

Training data ingestion

Mechanism flow

Source → Validation → Shard → Cache → Worker pool → Checkpoint

Service or method decision

- Cloud Storage shards — images, text and TFRecord
- BigQuery export — warehouse-managed tabular data
- Streaming ingest — continuous examples with later snapshot

Runnable configuration pattern

```
gcloud storage cp gs://$BUCKET/manifests/train-*.jsonl ./manifests/
python validate_manifest.py --schema schema.json --input manifests/
```

Metric: Input pipeline utilization

Formula/decision rule: utilization = data_wait_free_step_time / total_step_time

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Workers starve	tiny unshuffled files	shard and parallel-read
Corrupt examples	no validation	fail-fast schema checks
Nonrepeatable split	random at runtime	persist split manifest

Evidence chain

- manifest
- schema report
- shard counts
- input throughput
- checkpoint URI

Exam-style decision case

A vision job uses 12 million images and GPUs idle while workers open millions of tiny files.

Best decision: Create balanced shards, parallel reads and prefetch; measure accelerator utilization and input wait.

Related practical: Activity 10

TensorFlow and Keras training

Mechanism flow

tf.data → Preprocess → Keras model → Fit → Evaluate → SavedModel

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Service or method decision

- Sequential API — single-input linear stack
- Functional API — multi-input or branching graph
- Subclassing — custom behavior requiring code control

Runnable configuration pattern

```
ds =
tf.data.Dataset.from_tensor_slices((X,y)).shuffle(10000).batch(128).prefetch(tf.data
.AUTOTUNE)
model.fit(ds, epochs=20, callbacks=[early_stop])
```

Metric: Validation loss

Formula/decision rule: stop when val_loss fails to improve for patience epochs

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Overfit	training continues after val plateau	early stopping
Slow input	no prefetch	AUTOTUNE pipeline
Serving failure	custom op missing	export and load test

Evidence chain

- model summary
- training history
- validation metrics
- SavedModel signature
- reload test

Exam-style decision case

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

A tabular neural network improves training loss but validation loss rises after epoch 12.

Best decision: Use early stopping and regularization, then verify the exported signature with held-out examples.

Related practical: Activity 07

Vertex AI custom training

Mechanism flow

Package/container → Artifact Registry → CustomJob → Worker pool → GCS output → Model Registry

Service or method decision

- Prebuilt container — supported framework and conventional entry point
- Custom container — system packages or custom runtime
- Custom training package — Python package on managed image

Runnable configuration pattern

```
gcloud ai custom-jobs create --region=$REGION --display-name=churn-train \
+ --config=config/custom_job.yaml
```

Metric: Training cost per successful run

Formula/decision rule: cost_success = total_job_cost / successful_jobs

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Module not found	package omitted dependency	lock and build image
Permission denied	training SA lacks access	least-privilege roles
No artifact	wrong AIP_MODEL_DIR	write to managed output URI

Evidence chain

- image digest
- custom job spec
- service account
- Cloud Logging trace
- artifact checksum

Exam-style decision case

A sklearn pipeline requires a native library unavailable in Vertex prebuilt images.

Best decision: Use a custom container pinned by digest and test it locally before CustomJob submission.

Related practical: Activity 08

Hyperparameter tuning

Mechanism flow

Search space → Trials → Objective → Early stop → Best trial → Retrain

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Service or method decision

- Grid — small discrete exhaustive space
- Random — broad mixed space
- Bayesian/Vizier — expensive trials with guided search

Runnable configuration pattern

```
metric_spec: {val_auc: maximize}
parameters:
- {parameterId: learning_rate, doubleValueSpec: {minValue: 0.0001, maxValue: 0.1,
scaleType: UNIT_LOG_SCALE}}
```

Metric: Best objective by trial

Formula/decision rule: best_n = max(metric_1...metric_n)

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Wrong objective	metric name mismatch	emit exact metric
Budget waste	poor trials run fully	early stopping
Overfit to validation	too many searches	final untouched test

Evidence chain

- search-space spec
- trial logs
- objective series
- best parameters
- test-set confirmation

Exam-style decision case

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

A DNN training run costs $80; the team needs learning rate, depth and dropout optimization.

Best decision: Use Vizier/Bayesian search with log-scaled learning rate, early stopping and an untouched final test.

Related practical: Activity 09

Foundation model fine-tuning

Mechanism flow

Task fit → Data curation → Adapter/tune → Safety eval → Quality eval → Deploy

Service or method decision

- Prompt/RAG — knowledge or instruction gap
- Adapter tuning — task style with modest data
- Full fine-tune — rarely justified; high control and cost

Runnable configuration pattern

```
train.jsonl: {'input_text':'...','output_text':'...'}
validation.jsonl: {'input_text':'...','output_text':'...'}
max_duplicate_rate: 0.01
```

Metric: Quality vs training examples

Formula/decision rule: gain = tuned_score - base_score

Failure diagnosis

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Observable symptom	Likely mechanism	Control or verification
Catastrophic behavior	poor examples	curate and safety-gate
No factual freshness	tuning used for knowledge	use retrieval
Sensitive memorization	PII in examples	DLP and dedup

Evidence chain

- data approval
- base/tuned eval
- safety red-team
- training lineage
- deployment rollback

Exam-style decision case

A service needs consistent structured incident summaries but incident facts change every minute.

Best decision: Tune structure/style only; retrieve live facts and block memorization-prone sensitive examples.

Related practical: Activity 16

CPU, GPU and TPU selection

Mechanism flow

Profile → Ops → Memory → Precision → Throughput → Cost

Service or method decision

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- CPU — small classical models and irregular control flow
- GPU — broad deep-learning ecosystem
- TPU — large TensorFlow/JAX workloads optimized for matrix ops

Runnable configuration pattern

```
acceleratorType: NVIDIA_TESLA_T4
acceleratorCount: 2
machineSpec: {machineType: n1-standard-16}
replicaCount: 1
```

Metric: Examples per dollar

Formula/decision rule: efficiency = examples_per_second / hourly_cost

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
OOM	batch/model exceeds memory	gradient accumulation or larger device
Low utilization	input bottleneck	profile pipeline
Quota error	accelerator unavailable	reserve quota or alternate region

Evidence chain

- profiler trace
- memory peak
- throughput test
- quota record
- cost benchmark

Exam-style decision case

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

A transformer fits on one GPU but throughput misses the nightly window by 3x.

Best decision: Profile first; compare larger GPU and data-parallel options by examples per dollar and deadline.

Related practical: Activity 10

Distributed training

Mechanism flow

Sharded data → Workers → All-reduce → Optimizer → Checkpoint → Restart

Service or method decision

- Data parallel — replicate model; split batches
- Model parallel — split model too large for one device
- Parameter server — asynchronous large-scale pattern

Runnable configuration pattern

```
python -m torch.distributed.run --nproc_per_node=4 train.py \
+  --backend nccl --checkpoint_uri gs://$BUCKET/checkpoints
```

Metric: Scaling efficiency

Formula/decision rule: efficiency = speedup / worker_count

Failure diagnosis

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Observable symptom	Likely mechanism	Control or verification
Hang	rank/network mismatch	validate rendezvous and ports
Divergence	effective batch changed	scale learning rate
Lost work	no frequent checkpoint	checkpoint to durable GCS

Evidence chain

- worker-pool spec
- rank logs
- throughput curve
- checkpoint list
- restart test

Exam-style decision case

An ImageNet job must scale from 1 to 8 GPUs and survive preemptible worker loss.

Best decision: Use torchrun data parallelism, durable checkpoints and measure scaling efficiency before adding workers.

Related practical: Activity 10

Domain 4 — Serving and scaling models (~20%)

Topics 6–7: batch/online prediction, production systems.

Batch versus online inference

Mechanism flow

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Input contract → Preprocess → Model → Postprocess → Sink → SLA

Service or method decision

- Batch prediction — asynchronous high-volume jobs
- Online endpoint — interactive low-latency requests
- Streaming — continuous events with bounded delay

Runnable configuration pattern

```
gcloud ai batch-prediction-jobs create --region=$REGION --model=$MODEL_ID \
+ --jsonl-request=gcsSource.uris=gs://$BUCKET/requests.jsonl \
+ --jsonl-output=gcsDestination.outputUriPrefix=gs://$BUCKET/predictions/
```

Metric: Cost per 1,000 predictions

Formula/decision rule: cost_1k = total_serving_cost / predictions * 1000

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Timeout	online used for huge payload	batch job
Stale output	batch schedule late	freshness SLO
Duplicate score	retries not idempotent	request IDs

Evidence chain

- request schema
- job ID
- input/output counts
- latency distribution
- cost report

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Exam-style decision case

Marketing needs 50 million nightly scores; the UI needs only the latest stored score.

Best decision: Use batch prediction to a governed table; serve the stored score rather than 50 million online calls.

Related practical: Activity 12

Prebuilt and custom serving containers

Mechanism flow

Artifact → Container → Health route → Predict route → Endpoint → Probe

Service or method decision

- Prebuilt — supported framework and signature
- Custom — custom dependencies or server
- Cloud Run container — portable stateless HTTP serving

Runnable configuration pattern

```
FROM python:3.11-slim
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py model.joblib ./
CMD ["gunicorn","-b",":8080","app:app"]
```

Metric: Container startup time

Formula/decision rule: startup = ready_timestamp - instance_start

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
CrashLoop	bad command or missing file	local container smoke test
Unhealthy	wrong probe path/port	implement health route
ABI error	runtime mismatch	pin base image and wheels

Evidence chain

- Dockerfile
- image digest
- SBOM
- health response
- prediction contract test

Exam-style decision case

An XGBoost model requires custom preprocessing and a native package not in prebuilt serving images.

Best decision: Build a minimal custom image, pin the digest, generate an SBOM and probe health/predict routes.

Related practical: Activity 13

Model Registry and versioning

Mechanism flow

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Train → Evaluate → Register → Alias → Deploy → Retire

Service or method decision

- Version — immutable model artifact generation
- Alias — movable label such as champion
- Lineage — data/code/run relation

Runnable configuration pattern

```
model_labels = {'git_sha':GIT_SHA,'data_version':DATA_VER,'owner':'risk-ml'}
model_alias = 'candidate'
```

Metric: Rollback time

Formula/decision rule: MTTR = restore_time - incident_start

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Unknown model	manual file copied	registry-only deployment
Wrong alias	unreviewed promotion	approval gate
No rollback	old version deleted	retain validated prior version

Evidence chain

- registry version
- labels
- evaluation report
- promotion approval
- rollback drill

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Exam-style decision case

A model release must be traceable to code and data, and rollback must finish within 15 minutes.

Best decision: Register immutable versions with lineage labels; promote aliases only after gates and drill rollback.

Related practical: Activity 12

Canary and A/B rollout

Mechanism flow

Baseline → Candidate → Traffic split → Guardrails → Decision → Promote/Rollback

Service or method decision

- Canary — risk-limited technical validation
- A/B — compare user/business outcomes
- Shadow — observe candidate without affecting users

Runnable configuration pattern

```
trafficSplit:
  deployedModels:
    champion: 90
    candidate: 10
rollback_if: 'p95_ms > 300 or error_rate > 0.01'
```

Metric: Canary error rate

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Formula/decision rule: error_rate = failed_requests / total_requests

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Biased sample	non-random routing	stable random assignment
Slow rollback	manual process	automated guardrail
Metric dilution	aggregate traffic	per-version metrics

Evidence chain

- traffic config
- version-tagged logs
- guardrail dashboard
- decision record
- rollback event

Exam-style decision case

A new ranking model improves offline NDCG but has unknown tail latency at peak traffic.

Best decision: Start with shadow then 10% canary; gate on version-specific latency, errors and business metric.

Related practical: Activity 12

Inference preprocessing and postprocessing

Mechanism flow

Validate request → Transform → Predict → Calibrate → Business rule → Response

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Service or method decision

- In-container transform — tight coupling and low latency
- Pipeline transform — shared batch/online code
- Feature Store — central online feature retrieval

Runnable configuration pattern

```
assert request['schema_version']=='3'
x = transformer.transform([request['features']])
p = calibrator.predict_proba(x)[0,1]
return {'risk':float(p),'model_version':VERSION}
```

Metric: Training-serving skew

Formula/decision rule: skew = distance(train_feature, serving_feature)

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Shape error	schema changed	versioned contract
Probability drift	calibrator omitted	bundle postprocessor
Silent default	missing feature filled	reject or explicit missingness

Evidence chain

- JSON schema
- transform checksum
- golden request
- prediction diff test
- response version

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Exam-style decision case

The same raw input scores 0.22 offline and 0.71 online after a preprocessing refactor.

Best decision: Compare transformation artifacts and golden requests; deploy preprocessing with the model version.

Related practical: Activity 13

Public and private endpoints

Mechanism flow

Client → DNS → Identity → Network path → Endpoint → Audit

Service or method decision

- Public endpoint — internet path with IAM/auth controls
- Private Service Connect — private VPC consumption
- Cloud Run ingress — service-to-service serverless

Runnable configuration pattern

```
ingress: internal
authentication: iam
psc_network: projects/$PROJECT/global/networks/ml-vpc
log_requests: metadata_only
```

Metric: Unauthorized request rate

Formula/decision rule: denied / total requests

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
403	caller identity lacks invoke	grant narrow invoker
No route	PSC/DNS misconfigured	validate endpoint and DNS
Data leak	payload logs	metadata-only logging

Evidence chain

- network diagram
- IAM policy
- DNS resolution
- connectivity test
- audit log

Exam-style decision case

A bank model endpoint must be reachable from two VPCs and never expose public ingress.

Best decision: Use private connectivity, service identities and end-to-end connectivity plus audit tests.

Related practical: Activity 12

Autoscaling and serving hardware

Mechanism flow

Load test → Concurrency → Replica floor → Scale signal → Accelerator → SLO

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Service or method decision

- CPU autoscale — lightweight model and variable demand
- GPU endpoint — high-throughput deep learning
- Edge — connectivity or privacy requires local inference

Runnable configuration pattern

```
minReplicaCount: 2
maxReplicaCount: 12
targetCpuUtilization: 60
acceleratorType: NVIDIA_TESLA_T4
```

Metric: p95 latency by QPS

Formula/decision rule: p95 = percentile(latency_ms,95)

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Cold-start spike	scale from zero	warm floor
Queue growth	concurrency too high	capacity test
GPU idle	small batches	dynamic batching or CPU

Evidence chain

- load script
- QPS/latency curve
- replica timeline
- utilization
- capacity decision

Exam-style decision case

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

A vision API must handle 5–80 QPS with p95 below 300 ms and control GPU cost.

Best decision: Benchmark CPU/GPU, set a warm floor and autoscale from measured saturation—not guesses.

Related practical: Activity 13

Cloud Run and GKE serving

Mechanism flow

Image → Registry → Runtime → Service/Deployment → Autoscale → Observe

Service or method decision

- Cloud Run — stateless HTTP, rapid ops, bursty load
- GKE — custom networking, sidecars, accelerators or orchestration
- Vertex endpoint — managed ML deployment and monitoring

Runnable configuration pattern

```
gcloud run deploy churn-api --image=$IMAGE --region=$REGION \
+ --no-allow-unauthenticated --min=1 --max=20 --cpu=2 --memory=2Gi
```

Metric: Request latency

Formula/decision rule: latency = response_time - request_time

Failure diagnosis

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Observable symptom	Likely mechanism	Control or verification
Revision not ready	port or startup failure	read revision logs
OOM kill	memory limit	profile and right-size
5xx burst	downstream saturation	timeouts and circuit breaker

Evidence chain

- image digest
- deployment YAML/command
- revision ID
- load-test output
- logs and metrics

Exam-style decision case

A scikit-learn API is stateless and bursty; a future GPU version may need custom scheduling.

Best decision: Deploy CPU baseline to Cloud Run; use GKE only when its control requirements become real.

Related practical: Activity 13

Domain 5 — Automating and orchestrating ML pipelines (~18%)

Topic 8: MLOps, CI/CD and repeatable workflows.

Pipeline component contracts

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Mechanism flow

Ingest → Validate → Transform → Train → Evaluate → Deploy gate

Service or method decision

- Artifact input — versioned file/model output
- Parameter — small configuration value
- Metric — numeric gate and comparison

Runnable configuration pattern

```
@component
def evaluate(model: Input[Model], metrics: Output[Metrics]):
    score = run_eval(model.path)
    metrics.log_metric('auc', score)
```

Metric: Pipeline pass rate

Formula/decision rule: passes / scheduled_runs

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Hidden dependency	component reads global path	declare input
Cache poison	mutable URI	immutable artifact
Wrong deploy	gate not wired	explicit conditional

Evidence chain

- component spec
- input/output schema

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- compiled pipeline
- artifact URI
- gate result

Exam-style decision case

A pipeline retrains successfully but deploys a model evaluated on the previous run's dataset.

Best decision: Make data/model/metrics explicit artifacts and wire deployment to the current evaluation output.

Related practical: Activity 14

Vertex AI Pipelines and Kubeflow

Mechanism flow

Python DSL → Compile → Submit → Metadata → Cache → Schedule

Service or method decision

- Vertex AI Pipelines — managed KFP execution
- Kubeflow on GKE — cluster-level control
- Local runner — component unit testing only

Runnable configuration pattern

```
compiler.Compiler().compile(pipeline_func=training_pipeline,
package_path='pipeline.json')
```


© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

```
aiplatform.PipelineJob(display_name='churn', template_path='pipeline.json').run()
```

Metric: Cached step savings

Formula/decision rule: saved = uncached_duration - cached_duration

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Compile error	type mismatch	typed component contracts
Permission failure	pipeline SA incomplete	role matrix
Unexpected cache	inputs unchanged but code changed	version component/image

Evidence chain

- pipeline JSON
- service account
- run graph
- metadata lineage
- schedule config

Exam-style decision case

A weekly training pipeline must be reproducible, inspectable and rerun only changed components.

Best decision: Use Vertex AI Pipelines with versioned components, managed metadata and deliberate caching.

Related practical: Activity 14

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Cloud Composer and Airflow

Mechanism flow

DAG → Sensors → Data job → ML pipeline → SLA → Notification

Service or method decision

- Composer — cross-service scheduling and dependencies
- Vertex schedule — simple pipeline cadence
- Cloud Scheduler — single HTTP/event trigger

Runnable configuration pattern

```
with DAG('weekly_ml', schedule='0 2 * * 1', catchup=False) as dag:
    wait_data >> launch_dataflow >> run_vertex_pipeline >> verify_slo
```

Metric: DAG SLA misses

Formula/decision rule: miss_rate = missed_runs / scheduled_runs

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Backfill storm	catchup enabled	set catchup deliberately
Stuck sensor	polling resource	deferrable sensor
Duplicate run	non-idempotent task	run IDs and idempotency

Evidence chain

- DAG code
- run ID

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- task logs
- SLA miss
- notification ticket

Exam-style decision case

Training waits for three upstream data products and must notify owners when the freshness SLA fails.

Best decision: Use Composer for cross-team dependencies; keep ML logic inside the versioned Vertex pipeline.

Related practical: Activity 14

Ray on Vertex AI

Mechanism flow

Driver → Cluster → Tasks/actors → Object store → Autoscale → Result

Service or method decision

- Ray — distributed Python workloads and tuning
- Dataflow — Beam data processing
- CustomJob — bounded training job

Runnable configuration pattern

```
@ray.remote
def score_shard(uri):
```


© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

```
    return score(load(uri))
results = ray.get([score_shard.remote(u) for u in shard_uris])
```

Metric: Distributed task throughput

Formula/decision rule: throughput = completed_tasks / minute

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Object-store pressure	large returned objects	write artifacts externally
Task retries	non-idempotent side effect	idempotent tasks
Cluster idle	no teardown	job-scoped cluster policy

Evidence chain

- cluster spec
- Ray job ID
- task timeline
- object-store metrics
- result manifest

Exam-style decision case

A Python-native embedding job fans out 20,000 independent shards and aggregates a manifest.

Best decision: Use Ray tasks with external artifact writes, idempotent retries and explicit cluster teardown.

Related practical: Activity 14

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Training-serving preprocessing parity

Mechanism flow

Source schema → Transform graph → Training → Export → Serving → Golden test

Service or method decision

- tf.Transform — TensorFlow preprocessing graph
- Sklearn Pipeline — classical Python models
- Feature Store — central shared features

Runnable configuration pattern

```
expected = offline_pipeline.transform(golden)
actual = requests.post(endpoint, json=golden).json()['features']
np.testing.assert_allclose(actual, expected, rtol=1e-6)
```

Metric: Golden feature max error

Formula/decision rule: max_abs(actual-expected)

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Skew	two transform implementations	one exported graph
Schema mismatch	independent contracts	versioned schema
Silent rounding	different dtypes	assert tolerances

Evidence chain

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- transform artifact
- schema version
- golden input
- offline/online diff
- test result

Exam-style decision case

A category vocabulary updates in training, but the online service keeps last month's mapping.

Best decision: Version and deploy the transformation with the model; block rollout on golden parity tests.

Related practical: Activity 11

Retraining policy

Mechanism flow

Signal → Eligibility → Trigger → Train → Compare → Promote

Service or method decision

- Schedule — stable cadence and predictable data
- Drift trigger — distribution changes matter
- Performance trigger — labels arrive and business metric degrades

Runnable configuration pattern

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

```
trigger_if = {'psi_gt':0.2,'auc_drop_gt':0.03,'min_new_labels':10000}
cooldown_hours = 72
require_human_approval = True
```

Metric: AUC over time

Formula/decision rule: drop = baseline_auc - current_auc

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Retrain loop	no cooldown	minimum interval
Bad labels	freshness mistaken for quality	label validation
Auto-promote harm	single metric	multi-gate approval

Evidence chain

- trigger event
- data-quality report
- candidate/champion comparison
- approval
- promotion log

Exam-style decision case

Labels arrive weekly; data drift is common but only some drift reduces business performance.

Best decision: Trigger on performance with enough fresh labels; use drift as investigation, not automatic proof.

Related practical: Activity 16

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

CI/CD/CT with Cloud Build

Mechanism flow

Commit → Test → Build image → Compile pipeline → Evaluate → Promote

Service or method decision

- CI — unit, contract and security tests
- CD — deploy validated artifacts
- CT — retrain from governed triggers

Runnable configuration pattern

```
steps:
- name: python:3.11
  entrypoint: pytest
- name: gcr.io/cloud-builders/docker
  args: ['build','-t','$IMAGE','.']
images: ['$IMAGE']
```

Metric: Change failure rate

Formula/decision rule: failed_deployments / deployments

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Mutable latest	tag reused	deploy digest
Test gap	only code unit tests	data/model/contract tests
Privilege escalation	broad build SA	least-privilege build identity

Evidence chain

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- commit SHA
- test report
- image digest
- provenance/SBOM
- deployment approval

Exam-style decision case

A training image passes local tests but production deploys a different dependency set.

Best decision: Build once in Cloud Build, lock dependencies, attest the digest and deploy that exact artifact.

Related practical: Activity 15

Pipeline governance and reproducibility

Mechanism flow

Policy → Template → Execution → Metadata → Review → Retention

Service or method decision

- Org policy — prevent forbidden resource settings
- Template — approved repeatable architecture
- Policy-as-code — automated configuration gate

Runnable configuration pattern

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

```
required_labels = ['owner','data_classification','model_risk']
forbidden = ['public_ip','default_service_account']
retention_days = 365
```

Metric: Reproducible run rate

Formula/decision rule: reproduced_within_tolerance / audited_runs

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Missing owner	labels optional	admission check
Data unavailable	retention too short	align lineage retention
Environment drift	manual runtime	versioned container

Evidence chain

- policy result
- template version
- run metadata
- artifact retention
- audit replay

Exam-style decision case

An auditor selects a six-month-old prediction and asks which code, data and model produced it.

Best decision: Retain end-to-end lineage, immutable artifacts and policy evidence long enough for replay.

Related practical: Activity 15

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Domain 6 — Monitoring AI solutions (~13%)

Topics 6–8: testing, monitoring, troubleshooting and MLOps.

AI security and Model Armor

Mechanism flow

Input → Detect → Sanitize → Model → Output filter → Audit

Service or method decision

- Regex/DLP — known sensitive patterns
- Safety filters — harm categories
- Model Armor — managed prompt/response protection

Runnable configuration pattern

```
blocked_patterns = ['ignore previous instructions','BEGIN PRIVATE KEY']
scan_prompt = True
scan_response = True
log_payload = False
```

Metric: Attack success rate

Formula/decision rule: successful_attacks / red_team_attempts

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Indirect injection	retrieved document contains commands	treat context as data

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Exfiltration	model reveals secret	minimize context and output scan
Log exposure	payload retained	redaction and access control

Evidence chain

- threat model
- red-team corpus
- filter config
- blocked event
- incident review

Exam-style decision case

A grounded assistant retrieves a document that instructs the model to reveal system prompts.

Best decision: Separate instructions from untrusted context, scan inputs/outputs and test indirect injections.

Related practical: Activity 16

Responsible AI and bias monitoring

Mechanism flow

Outcome → Protected slices → Metric → Threshold → Review → Mitigation

Service or method decision

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Demographic parity — selection rates where appropriate
- Equal opportunity — true-positive rates
- Calibration — probabilities match outcomes

Runnable configuration pattern

```
monitor_slices = ['region','age_band']
max_tpr_gap = 0.05
min_sample = 200
escalate_to = 'model-risk-committee'
```

Metric: True-positive-rate gap

Formula/decision rule: max(TPR)-min(TPR)

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Small slice noise	insufficient N	confidence bounds/min sample
Proxy feature	correlated attribute	causal/domain review
Metric conflict	one fairness metric only	risk-specific metric set

Evidence chain

- slice definition
- metric with intervals
- threshold policy
- review minutes
- mitigation outcome

Exam-style decision case

Approval rates are stable overall but qualified applicants in one region see falling true-positive rate.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Best decision: Alert on slice TPR gap with sample safeguards and require model-risk review before threshold changes.

Related practical: Activity 16

Explainability and feature attribution

Mechanism flow

Prediction → Baseline → Attributions → Global summary → Slice review → Action

Service or method decision

- Local attribution — explain one prediction
- Global importance — aggregate model behavior
- Counterfactual — what feasible change alters outcome

Runnable configuration pattern

```
explanation_spec = {'method':'sampled_shapley','sampled_shapley_attribution':
{'path_count':20}}
```

Metric: Attribution drift

Formula/decision rule: distance(current_attributions, baseline_attributions)

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Causal claim	attribution mistaken for cause	state limitation
Unstable rank	correlated features	group and test stability
Sensitive proxy	unexpected importance	investigate governance

Evidence chain

- explanation config
- local plot
- global ranking
- slice comparison
- review action

Exam-style decision case

A credit model keeps AUC, but postcode suddenly becomes the top feature after a data release.

Best decision: Investigate attribution drift and proxy risk; do not accept stable accuracy as proof of safety.

Related practical: Activity 16

Data drift, skew and concept drift

Mechanism flow

Baseline → Current window → Statistic → Threshold → Diagnosis → Response

Service or method decision

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Training-serving skew — training vs serving pipeline mismatch
- Data drift — input distribution changes
- Concept drift — relationship between input and outcome changes

Runnable configuration pattern

```
psi = sum((current_pct - base_pct) * log(current_pct/base_pct))
alert_if = 'psi > 0.2 for 2 windows'
```

Metric: Population stability index

Formula/decision rule: PSI = sum((A-E)*ln(A/E))

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
False drift	seasonality	seasonal baseline
Skew	transform version differs	parity test
Concept drift	inputs stable but errors rise	label-based performance monitor

Evidence chain

- baseline version
- window definition
- drift statistic
- alert
- diagnosis and disposition

Exam-style decision case

A fraud feature distribution shifts every holiday, but loss rises only after a new payment channel launch.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Best decision: Use seasonal baselines; distinguish benign data drift from labeled concept drift before retraining.

Related practical: Activity 16

Gen AI evaluation in production

Mechanism flow

Sample → Safety scan → Groundedness → Task quality → Human audit → Feedback

Service or method decision

- Continuous eval — sample production traffic
- Synthetic red team — probe rare attacks
- Human audit — calibrate nuanced risk

Runnable configuration pattern

```
sample_rate = 0.02
metrics = ['groundedness','instruction_following','safety']
critical_violation_budget = 0
```

Metric: Grounded answer rate

Formula/decision rule: grounded_answers / evaluated_answers

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Privacy leak	raw prompts in eval set	redact and restrict

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Judge drift	judge model changed	version and calibrate
Silent regression	no release baseline	champion comparison

Evidence chain

- sampling policy
- redacted eval set
- judge version
- human agreement
- incident linkage

Exam-style decision case

A support assistant remains fluent but citations increasingly fail to support its claims.

Best decision: Continuously score groundedness, audit samples and block releases that regress against the champion.

Related practical: Activity 16

SLOs, logs and alerts

Mechanism flow

User journey → SLI → SLO → Error budget → Alert → Runbook

Service or method decision

- Availability — successful valid responses
- Latency — p95/p99 per request class

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Quality — validated prediction or grounded response

Runnable configuration pattern

```
slo:
  availability: 0.999
  p95_latency_ms: 300
  quality_floor: 0.82
alert_on_burn_rate: true
```

Metric: Error-budget burn

Formula/decision rule: burn = observed_error_rate / allowed_error_rate

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Alert storm	symptom alerts duplicate	burn-rate aggregation
No context	logs lack version	structured version fields
Green infra/red users	only CPU monitored	user-centric quality SLI

Evidence chain

- SLI query
- SLO policy
- version-tagged log
- burn alert
- runbook execution

Exam-style decision case

Infrastructure is healthy, yet users receive stale predictions for six hours after a failed batch.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Best decision: Add freshness and quality SLIs, not just infrastructure metrics; page on error-budget burn.

Related practical: Activity 16

Incident troubleshooting

Mechanism flow

Detect → Triage → Contain → Diagnose → Recover → Learn

Service or method decision

- Rollback — new version caused regression
- Scale — capacity saturation
- Disable feature — upstream corruption

Runnable configuration pattern

```
incident_id: ML-2026-0912
model_version: candidate-17
start: 14:05Z
containment: rollback_to_champion
owner: ml-platform
```

Metric: Mean time to recovery

Formula/decision rule: MTTR = sum(recovery_duration)/incident_count

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Slow triage	no version tags	structured logs
Repeat issue	no action owner	postmortem with due dates
Evidence loss	ephemeral logs	central retention

Evidence chain

- timeline
- impact estimate
- version and data IDs
- containment command
- postmortem actions

Exam-style decision case

p95 latency triples immediately after a new model version receives 20% traffic.

Best decision: Compare per-version metrics, roll back the candidate, preserve traces, then reproduce under load.

Related practical: Activity 16

Cost and performance optimization

Mechanism flow

Measure → Attribute → Profile → Change → Load test → Approve

Service or method decision

- Right-size — remove unused CPU/memory/accelerator

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Batching — amortize inference overhead
- Quantization — trade small quality loss for speed/cost

Runnable configuration pattern

```
baseline = {'p95_ms':280,'cost_1k':0.42,'auc':0.846}
candidate = {'p95_ms':170,'cost_1k':0.24,'auc':0.842}
max_auc_loss = 0.005
```

Metric: Cost per 1,000 predictions

Formula/decision rule: total_cost / prediction_count * 1000

Failure diagnosis

Observable symptom	Likely mechanism	Control or verification
Cheap but worse	quality not gated	multi-objective gate
Benchmark bias	toy load	production traffic shape
Hidden egress	cross-region path	co-locate services

Evidence chain

- billing labels
- profiler trace
- quality comparison
- load test
- architecture decision

Exam-style decision case

Quantization halves cost but reduces AUC by 0.004 and improves p95 latency by 110 ms.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Best decision: Accept only if the documented quality tolerance is 0.005 and slice/edge-case gates also pass.

Related practical: Activity 16

Detailed Activity Guides

The following sections reproduce the current learner-facing Activity guides. Each Activity folder also contains a same-basename PDF plus the mock data, scripts and configuration needed to complete it.

Activity 01 — Create a Safe Google Cloud Training Environment

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Create a Google account, Cloud project, billing budget, IAM boundary and Cloud Shell workspace.

Deliverables

- budget-policy.json
- check_environment.sh
- iam-role-matrix.csv
- A completed evidence/ folder with decision notes and verification output

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: a Google account you control and trainer approval to create an isolated training project.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Open a private browser window and go to https://accounts.google.com/signup

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Open a private browser window and go to https://accounts.google.com/signup. Create a dedicated training Google Account or sign in with an approved existing account. Complete identity and recovery checks; do not use a shared password.

Check: Record the observable result or explain the error before continuing.

Step 2 — Go to https://console.cloud.google.com/

Go to https://console.cloud.google.com/. Accept the Google Cloud terms when prompted. If the free-trial offer is available and appropriate, review the current terms before entering payment details; free-tier offers can change.

Check: Record the observable result or explain the error before continuing.

Step 3 — Open the project selector, choose New Project, name it pmle-training-<initials>, record the generated project ID, and select the correct organisation or No organisation only if your trainer approves

Open the project selector, choose New Project, name it pmle-training-<initials>, record the generated project ID, and select the correct organisation or No organisation only if your trainer approves.

Check: Record the observable result or explain the error before continuing.

Step 4 — Open Billing and link the training project to the authorised billing account

Open Billing and link the training project to the authorised billing account. Billing enables paid services; it does not make all usage free.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 5 — Open Billing > Budgets & alerts, create a small training budget, and add alert thresholds at 50%, 90% and 100%

Open Billing > Budgets & alerts, create a small training budget, and add alert thresholds at 50%, 90% and 100%. Billing alerts do not automatically stop resources.

Check: Record the observable result or explain the error before continuing.

Step 6 — Open IAM & Admin > IAM

Open IAM & Admin > IAM. Confirm your user has only the roles required for training. Do not grant Owner to classmates or service accounts.

Check: Record the observable result or explain the error before continuing.

Step 7 — Open APIs & Services > Library and enable Vertex AI, BigQuery, Cloud Storage, Artifact Registry, Cloud Build and Cloud Run APIs

Open APIs & Services > Library and enable Vertex AI, BigQuery, Cloud Storage, Artifact Registry, Cloud Build and Cloud Run APIs. Record any enablement error before changing roles.

Check: Record the observable result or explain the error before continuing.

Step 8 — Activate Cloud Shell from the console header

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Activate Cloud Shell from the console header. Cloud Shell already includes the Google Cloud CLI and uses your signed-in identity.

```
gcloud auth list
```

Check: Record the observable result or explain the error before continuing.

Step 9 — Set and verify project and region variables for this shell session

Set and verify project and region variables for this shell session.

```
export PROJECT_ID='<PROJECT_ID>' REGION='asia-southeast1'
gcloud config set project "$PROJECT_ID"
gcloud config set ai/region "$REGION"
```

Check: Record the observable result or explain the error before continuing.

Step 10 — Run the supplied environment checker and save its output as evidence

Run the supplied environment checker and save its output as evidence.

```
bash check_environment.sh | tee evidence/activity-01-environment.txt
```

Check: Record the observable result or explain the error before continuing.

Step 11 — Review iam-role-matrix.csv

Review iam-role-matrix.csv. For each activity, grant the narrow predefined role only when needed; avoid service-account keys.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 12 — Record the project ID, billing account alias, budget name, enabled APIs and IAM decision in your activity evidence

Record the project ID, billing account alias, budget name, enabled APIs and IAM decision in your activity evidence. Do not record payment details.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version	Compare file hashes, pinned versions and

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

	drifted	parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 02 — Draft an ML Architecture Decision Record

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Convert business volume, latency, security and explainability needs into a service decision.

Deliverables

- requirements.csv
- architecture_decision.py
- expected_decision.json
- A completed evidence/ folder with decision notes and verification output

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Open requirements.csv and identify the business volume, freshness, latency, availability, privacy and explainability requirements

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Open requirements.csv and identify the business volume, freshness, latency, availability, privacy and explainability requirements.

Check: Record the observable result or explain the error before continuing.

Step 2 — Run the architecture evaluator to score BigQuery ML, AutoML and custom training options

Run the architecture evaluator to score BigQuery ML, AutoML and custom training options.

```
python3 architecture_decision.py --requirements requirements.csv --out decision.json
```

Check: Record the observable result or explain the error before continuing.

Step 3 — Compare decision.json with expected_decision.json; investigate any difference instead of overwriting it

Compare decision.json with expected_decision.json; investigate any difference instead of overwriting it.

```
python3 -m json.tool decision.json
```

Check: Record the observable result or explain the error before continuing.

Step 4 — Draft an Architecture Decision Record with context, options, measurable drivers, selected service, rejected options and consequences

Draft an Architecture Decision Record with context, options, measurable drivers, selected service, rejected options and consequences.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 5 — Add a capacity calculation for peak QPS, 30% headroom and data growth over two years

Add a capacity calculation for peak QPS, 30% headroom and data growth over two years.

Check: Record the observable result or explain the error before continuing.

Step 6 — Add security controls for identity, encryption, network path, PII and audit logging

Add security controls for identity, encryption, network path, PII and audit logging.

Check: Record the observable result or explain the error before continuing.

Step 7 — Add testable acceptance criteria: p95 latency, quality floor, error rate, cost ceiling and rollback time

Add testable acceptance criteria: p95 latency, quality floor, error rate, cost ceiling and rollback time.

Check: Record the observable result or explain the error before continuing.

Step 8 — Peer-review against the documented acceptance criteria and save the review notes

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Peer-review against the documented acceptance criteria and save the review notes.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 03 — Train and Evaluate a BigQuery ML Churn Model

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Run SQL-defined logistic regression, evaluation and batch prediction on mock customer data.

Deliverables

- customers.csv
- bigquery_ml.sql
- evaluate_metrics.py
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Inspect customers.csv; identify the label, entity key and any feature that could leak the outcome

Inspect customers.csv; identify the label, entity key and any feature that could leak the outcome.

```
python3 - <<'PY'
import pandas as pd
df=pd.read_csv('customers.csv'); print(df.head()); print(df.isna().sum())
PY
```


© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 2 — Open bigquery_ml.sql and replace <PROJECT_ID> and <DATASET> without changing the point-in-time split

Open bigquery_ml.sql and replace <PROJECT_ID> and <DATASET> without changing the point-in-time split.

Check: Record the observable result or explain the error before continuing.

Step 3 — Create the dataset in the same approved region as the training project

Create the dataset in the same approved region as the training project.

```
bq --location=asia-southeast1 mk --dataset "$PROJECT_ID:pmle_lab"
```

Check: Record the observable result or explain the error before continuing.

Step 4 — Load the CSV using the declared schema or autodetect only after inspecting types

Load the CSV using the declared schema or autodetect only after inspecting types.

```
bq load --autodetect --source_format=CSV pmle_lab.customers customers.csv
```

Check: Record the observable result or explain the error before continuing.

Step 5 — Run the CREATE MODEL statement and record the BigQuery job ID and bytes processed

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Run the CREATE MODEL statement and record the BigQuery job ID and bytes processed.

```
bq query --use_legacy_sql=false < bigquery_ml.sql
```

Check: Record the observable result or explain the error before continuing.

Step 6 — Query ML.EVALUATE and save accuracy, precision, recall, F1, log loss and ROC AUC

Query ML.EVALUATE and save accuracy, precision, recall, F1, log loss and ROC AUC.

Check: Record the observable result or explain the error before continuing.

Step 7 — Run evaluate_metrics.py to compare the supplied confusion matrices at three thresholds

Run evaluate_metrics.py to compare the supplied confusion matrices at three thresholds.

```
python3 evaluate_metrics.py
```

Check: Record the observable result or explain the error before continuing.

Step 8 — Choose a threshold using the stated false-negative cost, not accuracy alone, and document the rationale

Choose a threshold using the stated false-negative cost, not accuracy alone, and document the rationale.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 9 — Run ML.PREDICT for the holdout rows and check that no post-outcome feature enters the model

Run ML.PREDICT for the holdout rows and check that no post-outcome feature enters the model.

Check: Record the observable result or explain the error before continuing.

Step 10 — Delete the model and dataset during cleanup if the trainer does not require them for a later activity

Delete the model and dataset during cleanup if the trainer does not require them for a later activity.

```
bq rm -f -m pmle_lab.churn_model
```

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 04 — Design an AutoML Tabular Experiment

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Objective

Create a dataset specification, split plan, budget and evaluation slice gate.

Deliverables

- automl_dataset.csv
- automl_config.yaml
- validate_splits.py
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Profile automl_dataset.csv and confirm the target has no nulls and each split is represented

Profile automl_dataset.csv and confirm the target has no nulls and each split is represented.

```
python3 validate_splits.py automl_dataset.csv
```

Check: Record the observable result or explain the error before continuing.

Step 2 — Review automl_config.yaml: objective, target, optimization metric, training budget and encryption setting

Review automl_config.yaml: objective, target, optimization metric, training budget and encryption setting.

Check: Record the observable result or explain the error before continuing.

Step 3 — Check whether a random split is valid for the event timestamps

Check whether a random split is valid for the event timestamps. Replace it with chronological split logic if future leakage is possible.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 4 — Define evaluation slices for region, channel and customer tenure

Define evaluation slices for region, channel and customer tenure. Set a minimum sample size for each slice.

Check: Record the observable result or explain the error before continuing.

Step 5 — Estimate training cost from node-hours before starting any AutoML job

Estimate training cost from node-hours before starting any AutoML job. Obtain trainer approval for paid execution.

Check: Record the observable result or explain the error before continuing.

Step 6 — In Vertex AI, create a tabular dataset only if approved; otherwise treat the configuration as a dry-run design exercise

In Vertex AI, create a tabular dataset only if approved; otherwise treat the configuration as a dry-run design exercise.

Check: Record the observable result or explain the error before continuing.

Step 7 — Configure the target column and split column, then compare the console summary with the YAML design

Configure the target column and split column, then compare the console summary with the YAML design.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 8 — After training, record model evaluation, feature attributions, slice results and training budget used

After training, record model evaluation, feature attributions, slice results and training budget used.

Check: Record the observable result or explain the error before continuing.

Step 9 — Write a go/no-go decision

Write a go/no-go decision. A high aggregate score cannot override a failed critical slice.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 05 — Compare Pretrained Google Cloud AI APIs

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Prepare Vision, Document AI and Natural Language requests and interpret mock responses.

Deliverables

- vision_request.json
- mock_vision_response.json
- api_selector.py
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Inspect vision_request.json and identify the feature type, image source and maximum results

Inspect vision_request.json and identify the feature type, image source and maximum results.

```
python3 -m json.tool vision_request.json
```

Check: Record the observable result or explain the error before continuing.

Step 2 — Inspect mock_vision_response.json; list detected labels, confidence scores and the response field holding errors

Inspect mock_vision_response.json; list detected labels, confidence scores and the response field holding errors.

Check: Record the observable result or explain the error before continuing.

Step 3 — Run the selector for three scenarios and record the recommended managed API or custom-model path

Run the selector for three scenarios and record the recommended managed API or custom-model path.

```
python3 api_selector.py
```


© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 4 — Prepare an authenticated REST command using gcloud auth print-access-token; do not store the token

Prepare an authenticated REST command using gcloud auth print-access-token; do not store the token.

Check: Record the observable result or explain the error before continuing.

Step 5 — If the trainer approves live use, enable the chosen API and send only the supplied non-sensitive sample

If the trainer approves live use, enable the chosen API and send only the supplied non-sensitive sample.

Check: Record the observable result or explain the error before continuing.

Step 6 — Interpret 403, 429 and low-confidence responses separately: authorization, quota and model fit are different problems

Interpret 403, 429 and low-confidence responses separately: authorization, quota and model fit are different problems.

Check: Record the observable result or explain the error before continuing.

Step 7 — Design a confidence-based human-review queue and calculate the expected review rate from the mock response set

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Design a confidence-based human-review queue and calculate the expected review rate from the mock response set.

Check: Record the observable result or explain the error before continuing.

Step 8 — Save the request, redacted response, operation ID, cost estimate and decision note

Save the request, redacted response, operation ID, cost estimate and decision note.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 06 — Build a Data Quality and Feature Pipeline

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Validate schema, repair missing values, encode features and preserve a training manifest.

Deliverables

- transactions.csv

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- build_features.py
- feature_schema.json
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Step 1 — Profile transactions.csv for types, nulls, duplicates, impossible values and label balance

Profile transactions.csv for types, nulls, duplicates, impossible values and label balance.

```
python3 build_features.py --input transactions.csv --profile-only
```

Check: Record the observable result or explain the error before continuing.

Step 2 — Read feature_schema.json and confirm each field's type, allowed range and missing-value policy

Read feature_schema.json and confirm each field's type, allowed range and missing-value policy.

Check: Record the observable result or explain the error before continuing.

Step 3 — Run the feature builder to create a deterministic train/validation/test split and manifest

Run the feature builder to create a deterministic train/validation/test split and manifest.

```
python3 build_features.py --input transactions.csv --out-dir output
```

Check: Record the observable result or explain the error before continuing.

Step 4 — Inspect the rejected-row report

Inspect the rejected-row report. Correct source data rules rather than silently dropping all failures.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 5 — Check that category encoding handles unknown values and that fitted statistics use training rows only

Check that category encoding handles unknown values and that fitted statistics use training rows only.

Check: Record the observable result or explain the error before continuing.

Step 6 — Verify entity IDs do not appear as predictive numeric features

Verify entity IDs do not appear as predictive numeric features.

Check: Record the observable result or explain the error before continuing.

Step 7 — Compare split timestamps and confirm no future event contributes to an earlier example

Compare split timestamps and confirm no future event contributes to an earlier example.

Check: Record the observable result or explain the error before continuing.

Step 8 — Record row-count reconciliation, schema signature and output checksums

Record row-count reconciliation, schema signature and output checksums.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 07 — Train a TensorFlow/Keras Model

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Use tf.data, a Keras network, early stopping and SavedModel verification.

Deliverables

- train_tensorflow.py
- requirements.txt
- expected_history.csv
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Create a clean virtual environment and install the pinned requirements

Create a clean virtual environment and install the pinned requirements.

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Check: Record the observable result or explain the error before continuing.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Step 2 — Open train_tensorflow.py and locate shuffle, batch, prefetch, early stopping and the export signature

Open train_tensorflow.py and locate shuffle, batch, prefetch, early stopping and the export signature.

Check: Record the observable result or explain the error before continuing.

Step 3 — Run the offline training script on the generated mock dataset

Run the offline training script on the generated mock dataset.

```
python3 train_tensorflow.py --epochs 20 --output output/model
```

Check: Record the observable result or explain the error before continuing.

Step 4 — Compare training and validation loss with expected_history.csv and identify the first overfitting epoch

Compare training and validation loss with expected_history.csv and identify the first overfitting epoch.

Check: Record the observable result or explain the error before continuing.

Step 5 — Reload the SavedModel and score the included golden example

Reload the SavedModel and score the included golden example.

```
python3 train_tensorflow.py --verify-only --output output/model
```


© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 6 — Change batch size once, rerun, and record throughput and validation quality rather than selecting by speed alone

Change batch size once, rerun, and record throughput and validation quality rather than selecting by speed alone.

Check: Record the observable result or explain the error before continuing.

Step 7 — Save the model summary, history, evaluation metrics and signature output

Save the model summary, history, evaluation metrics and signature output.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 08 — Package a Vertex AI Custom Training Job

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Build a reproducible training package and container with managed output paths.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Deliverables

- train.py
- Dockerfile
- custom_job.yaml
- requirements.txt
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Detailed step-by-step guide

Step 1 — Inspect train.py, requirements.txt, Dockerfile and custom_job.yaml; trace each file to the managed job lifecycle

Inspect train.py, requirements.txt, Dockerfile and custom_job.yaml; trace each file to the managed job lifecycle.

Check: Record the observable result or explain the error before continuing.

Step 2 — Run the training entry point locally and confirm it writes to the AIP_MODEL_DIR fallback

Run the training entry point locally and confirm it writes to the AIP_MODEL_DIR fallback.

```
python3 train.py --data-uri mock://customers --epochs 3
```

Check: Record the observable result or explain the error before continuing.

Step 3 — Build the container locally with a versioned tag; do not use latest as deployment evidence

Build the container locally with a versioned tag; do not use latest as deployment evidence.

```
docker build -t pmle-train:v8 .
```

Check: Record the observable result or explain the error before continuing.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Step 4 — Run the container with a mounted output directory and verify the model artifact appears

Run the container with a mounted output directory and verify the model artifact appears.

```
docker run --rm -v "$PWD/output:/model" -e AIP_MODEL_DIR=/model pmle-train:v8
```

Check: Record the observable result or explain the error before continuing.

Step 5 — Replace project and Artifact Registry placeholders only if a live push is approved

Replace project and Artifact Registry placeholders only if a live push is approved.

Check: Record the observable result or explain the error before continuing.

Step 6 — Validate the worker-pool machine type, service account and artifact output permissions in custom_job.yaml

Validate the worker-pool machine type, service account and artifact output permissions in custom_job.yaml.

Check: Record the observable result or explain the error before continuing.

Step 7 — Submit the job only with trainer approval; record job ID, image digest and log link

Submit the job only with trainer approval; record job ID, image digest and log link.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 8 — Diagnose a supplied missing-dependency failure and propose a pinned dependency fix

Diagnose a supplied missing-dependency failure and propose a pinned dependency fix.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from	Data, dependency or	Compare file hashes,

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

expected	configuration version drifted	pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 09 — Analyze Vertex AI Hyperparameter Trials

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Define a Vizier-style search space and select a trial using quality and cost evidence.

Deliverables

- trials.csv
- tuning_config.yaml
- select_trial.py
- A completed evidence/ folder with decision notes and verification output

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Review tuning_config.yaml; identify search distributions, objective direction, maximum trials and parallel trials

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Review tuning_config.yaml; identify search distributions, objective direction, maximum trials and parallel trials.

Check: Record the observable result or explain the error before continuing.

Step 2 — Inspect trials.csv for status, objective, duration and cost

Inspect trials.csv for status, objective, duration and cost. Exclude failed trials from the quality ranking but retain them for reliability analysis.

Check: Record the observable result or explain the error before continuing.

Step 3 — Run the selector with the default quality floor and cost ceiling

Run the selector with the default quality floor and cost ceiling.

```
python3 select_trial.py --input trials.csv --min-auc 0.82 --max-cost 25
```

Check: Record the observable result or explain the error before continuing.

Step 4 — Calculate marginal quality gain from the third to the fifth completed trial

Calculate marginal quality gain from the third to the fifth completed trial.

Check: Record the observable result or explain the error before continuing.

Step 5 — Identify whether the best validation trial is likely over-searched and state why an untouched test set is required

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Identify whether the best validation trial is likely over-searched and state why an untouched test set is required.

Check: Record the observable result or explain the error before continuing.

Step 6 — Change the cost ceiling and explain the selected-trial change using evidence

Change the cost ceiling and explain the selected-trial change using evidence.

Check: Record the observable result or explain the error before continuing.

Step 7 — Draft a final retraining configuration using the selected parameters and no tuning-only callbacks

Draft a final retraining configuration using the selected parameters and no tuning-only callbacks.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 10 — Plan Distributed PyTorch GPU Training

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Objective

Configure torchrun-style workers, checkpoints and scaling-efficiency diagnostics.

Deliverables

- train_distributed.py
- Dockerfile
- worker_pool.yaml
- scaling_results.csv
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Read worker_pool.yaml and map chief, worker replicas, machine types, accelerator count and checkpoint URI

Read worker_pool.yaml and map chief, worker replicas, machine types, accelerator count and checkpoint URI.

Check: Record the observable result or explain the error before continuing.

Step 2 — Inspect train_distributed.py; locate process-group initialization, distributed sampler, all-reduce behavior and checkpoint save

Inspect train_distributed.py; locate process-group initialization, distributed sampler, all-reduce behavior and checkpoint save.

Check: Record the observable result or explain the error before continuing.

Step 3 — Run the local two-process smoke test without GPUs

Run the local two-process smoke test without GPUs.

```
python3 -m torch.distributed.run --standalone --nproc_per_node=2
train_distributed.py --epochs 2
```


© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 4 — Inspect scaling_results.csv and calculate speedup and scaling efficiency for 1, 2, 4 and 8 workers

Inspect scaling_results.csv and calculate speedup and scaling efficiency for 1, 2, 4 and 8 workers.

Check: Record the observable result or explain the error before continuing.

Step 5 — Explain why 8 workers may reduce efficiency even while throughput increases

Explain why 8 workers may reduce efficiency even while throughput increases.

Check: Record the observable result or explain the error before continuing.

Step 6 — Design a checkpoint interval from the observed failure rate and checkpoint write time

Design a checkpoint interval from the observed failure rate and checkpoint write time.

Check: Record the observable result or explain the error before continuing.

Step 7 — Build the custom container only if Docker is available; record the image digest

Build the custom container only if Docker is available; record the image digest.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

```
docker build -t pmle-torchrun:v8 .
```

Check: Record the observable result or explain the error before continuing.

Step 8 — Propose a GPU/TPU/CPU choice using examples per dollar, framework support and quota

Propose a GPU/TPU/CPU choice using examples per dollar, framework support and quota.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or	Recreate only the training

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

	endpoint uses another location	resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 11 — Verify Feature Store Point-in-Time Parity

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Model offline/online feature definitions and detect stale or leaked features.

Deliverables

- offline_features.csv
- online_features.csv
- check_parity.py

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- feature_view.yaml
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Step 1 — Inspect feature_view.yaml and identify the entity key, feature names, online TTL and source timestamp

Inspect feature_view.yaml and identify the entity key, feature names, online TTL and source timestamp.

Check: Record the observable result or explain the error before continuing.

Step 2 — Run the parity checker on the offline and online snapshots

Run the parity checker on the offline and online snapshots.

```
python3 check_parity.py --offline offline_features.csv --online online_features.csv
```

Check: Record the observable result or explain the error before continuing.

Step 3 — Separate point-in-time leakage from stale online values; they require different fixes

Separate point-in-time leakage from stale online values; they require different fixes.

Check: Record the observable result or explain the error before continuing.

Step 4 — Trace a mismatched entity through key normalization and timestamp selection

Trace a mismatched entity through key normalization and timestamp selection.

Check: Record the observable result or explain the error before continuing.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Step 5 — Set a freshness SLO and specify the monitoring window and alert persistence

Set a freshness SLO and specify the monitoring window and alert persistence.

Check: Record the observable result or explain the error before continuing.

Step 6 — Create a golden entity lookup and assert identical transformed values for batch and online paths

Create a golden entity lookup and assert identical transformed values for batch and online paths.

Check: Record the observable result or explain the error before continuing.

Step 7 — Record mismatches, root cause, remediation and the feature-definition version

Record mismatches, root cause, remediation and the feature-definition version.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 12 — Register and Canary a Model Version

Course: Google Professional Machine Learning Engineer Training

Course code: C740

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Create registry metadata, traffic split, guardrails and rollback evidence.

Deliverables

- model_manifest.json
- deployment.yaml
- canary_metrics.csv
- evaluate_canary.py
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Inspect model_manifest.json; verify model version, code SHA, data version, metrics and owner are complete

Inspect model_manifest.json; verify model version, code SHA, data version, metrics and owner are complete.

```
python3 -m json.tool model_manifest.json
```

Check: Record the observable result or explain the error before continuing.

Step 2 — Review deployment.yaml; locate the champion/candidate traffic split and rollback guardrails

Review deployment.yaml; locate the champion/candidate traffic split and rollback guardrails.

Check: Record the observable result or explain the error before continuing.

Step 3 — Run the canary evaluator against canary_metrics.csv

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Run the canary evaluator against canary_metrics.csv.

```
python3 evaluate_canary.py --metrics canary_metrics.csv --config deployment.yaml
```

Check: Record the observable result or explain the error before continuing.

Step 4 — Explain whether the candidate should be promoted, held or rolled back using per-version metrics

Explain whether the candidate should be promoted, held or rolled back using per-version metrics.

Check: Record the observable result or explain the error before continuing.

Step 5 — Change traffic to 25% only after the 10% stage passes; never overwrite prior stage evidence

Change traffic to 25% only after the 10% stage passes; never overwrite prior stage evidence.

Check: Record the observable result or explain the error before continuing.

Step 6 — Simulate a latency breach and confirm the evaluator returns a rollback decision

Simulate a latency breach and confirm the evaluator returns a rollback decision.

```
python3 evaluate_canary.py --metrics canary_metrics.csv --config deployment.yaml --
simulate-breach
```


© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 7 — Record registry ID, immutable artifact checksum, traffic split, decision and rollback test

Record registry ID, immutable artifact checksum, traffic split, decision and rollback test.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from	Data, dependency or	Compare file hashes,

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

expected	configuration version drifted	pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 13 — Serve a Model on Cloud Run

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Package a Flask prediction API, test its contract and analyze load-test results.

Deliverables

- app.py
- Dockerfile
- requirements.txt
- test_api.py
- load_results.csv
- A completed evidence/ folder with decision notes and verification output

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Create a virtual environment and install pinned dependencies

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Create a virtual environment and install pinned dependencies.

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Check: Record the observable result or explain the error before continuing.

Step 2 — Run the Flask service locally and verify /health before using /predict

Run the Flask service locally and verify /health before using /predict.

```
python3 app.py
```

Check: Record the observable result or explain the error before continuing.

Step 3 — In another terminal, run the API contract tests

In another terminal, run the API contract tests.

```
python3 test_api.py --base-url http://127.0.0.1:8080
```

Check: Record the observable result or explain the error before continuing.

Step 4 — Build the container and run the same health and prediction tests against it

Build the container and run the same health and prediction tests against it.

```
docker build -t pmle-api:v8 .
```


© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

```
docker run --rm -p 8080:8080 pmle-api:v8
```

Check: Record the observable result or explain the error before continuing.

Step 5 — Inspect load_results.csv; plot QPS against p95 latency and identify saturation

Inspect load_results.csv; plot QPS against p95 latency and identify saturation.

Check: Record the observable result or explain the error before continuing.

Step 6 — Draft a Cloud Run deployment with authenticated ingress, min/max instances and region

Draft a Cloud Run deployment with authenticated ingress, min/max instances and region.

Check: Record the observable result or explain the error before continuing.

Step 7 — Deploy only if approved, then record revision ID and image digest

Deploy only if approved, then record revision ID and image digest.

Check: Record the observable result or explain the error before continuing.

Step 8 — Test one invalid schema and confirm the service returns a controlled 4xx response without secrets

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Test one invalid schema and confirm the service returns a controlled 4xx response without secrets.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 14 — Compile a Vertex AI Pipeline

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Connect validate, transform, train, evaluate and conditional deployment components.

Deliverables

- pipeline.py
- pipeline_spec.json
- component_contracts.yaml
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Read component_contracts.yaml and identify explicit inputs, outputs, artifacts and metrics for every component

Read component_contracts.yaml and identify explicit inputs, outputs, artifacts and metrics for every component.

Check: Record the observable result or explain the error before continuing.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Step 2 — Open pipeline.py; trace validate, transform, train, evaluate and conditional deploy dependencies

Open pipeline.py; trace validate, transform, train, evaluate and conditional deploy dependencies.

Check: Record the observable result or explain the error before continuing.

Step 3 — Run the local compiler to produce compiled_pipeline.json

Run the local compiler to produce compiled_pipeline.json.

```
python3 pipeline.py --compile compiled_pipeline.json
```

Check: Record the observable result or explain the error before continuing.

Step 4 — Compare the compiled graph with pipeline_spec.json; confirm deployment consumes the current evaluation artifact

Compare the compiled graph with pipeline_spec.json; confirm deployment consumes the current evaluation artifact.

Check: Record the observable result or explain the error before continuing.

Step 5 — Change the quality threshold, recompile and inspect the parameter diff

Change the quality threshold, recompile and inspect the parameter diff.

Check: Record the observable result or explain the error before continuing.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Step 6 — Explain which components may be cached and which must rerun after a data-version change

Explain which components may be cached and which must rerun after a data-version change.

Check: Record the observable result or explain the error before continuing.

Step 7 — Design the pipeline service-account role matrix and reject default broad identities

Design the pipeline service-account role matrix and reject default broad identities.

Check: Record the observable result or explain the error before continuing.

Step 8 — Record compiled checksum, component versions and expected lineage path

Record compiled checksum, component versions and expected lineage path.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 15 — Design an ML CI/CD/CT Release

Course: Google Professional Machine Learning Engineer Training

Course code: C740

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Create Cloud Build gates, artifact provenance and a model card release record.

Deliverables

- cloudbuild.yaml
- release_gate.py
- model_card.md
- A completed evidence/ folder with decision notes and verification output

Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Inspect cloudbuild.yaml; classify each step as CI, CD or CT and identify the artifact passed forward

Inspect cloudbuild.yaml; classify each step as CI, CD or CT and identify the artifact passed forward.

Check: Record the observable result or explain the error before continuing.

Step 2 — Run the release gate on the supplied model metrics

Run the release gate on the supplied model metrics.

```
python3 release_gate.py --auc 0.846 --p95-ms 170 --fairness-gap 0.03
```

Check: Record the observable result or explain the error before continuing.

Step 3 — Run a failing case and confirm promotion is blocked with a clear reason

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Run a failing case and confirm promotion is blocked with a clear reason.

```
python3 release_gate.py --auc 0.79 --p95-ms 410 --fairness-gap 0.09
```

Check: Record the observable result or explain the error before continuing.

Step 4 — Open model_card.md and complete intended use, out-of-scope use, data, metrics, slices, limitations and owner

Open model_card.md and complete intended use, out-of-scope use, data, metrics, slices, limitations and owner.

Check: Record the observable result or explain the error before continuing.

Step 5 — Add dependency lock, unit test, data-contract test, container scan and provenance to the release evidence

Add dependency lock, unit test, data-contract test, container scan and provenance to the release evidence.

Check: Record the observable result or explain the error before continuing.

Step 6 — Specify which identity Cloud Build uses and the narrow roles it needs

Specify which identity Cloud Build uses and the narrow roles it needs.

Check: Record the observable result or explain the error before continuing.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Step 7 — Create a rollback plan referencing the previous registry version and its tested endpoint configuration

Create a rollback plan referencing the previous registry version and its tested endpoint configuration.

Check: Record the observable result or explain the error before continuing.

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

Activity 16 — Investigate Drift, Bias and an ML Incident

Course: Google Professional Machine Learning Engineer Training

Course code: C740

Version: v8.0 · 12 September 2026

Evidence: activity artifacts and acceptance checks

Objective

Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.

Deliverables

- baseline.csv
- current.csv
- monitor.py
- incident_template.md
- A completed evidence/ folder with decision notes and verification output

Prerequisites

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables,

screenshots without secrets, and a short decision note. Use filenames that include the activity

number and date. Evidence is part of the learning outcome, not an optional extra.

Detailed step-by-step guide

Step 1 — Inspect baseline.csv and current.csv; identify the monitored feature, outcome, prediction, group and timestamps

Inspect baseline.csv and current.csv; identify the monitored feature, outcome, prediction, group and timestamps.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Check: Record the observable result or explain the error before continuing.

Step 2 — Run the monitor to calculate PSI, accuracy, group true-positive rates and p95 latency

Run the monitor to calculate PSI, accuracy, group true-positive rates and p95 latency.

```
python3 monitor.py --baseline baseline.csv --current current.csv --out
monitoring_report.json
```

Check: Record the observable result or explain the error before continuing.

Step 3 — Classify each signal as training-serving skew, data drift, concept drift, bias risk or service degradation

Classify each signal as training-serving skew, data drift, concept drift, bias risk or service degradation.

Check: Record the observable result or explain the error before continuing.

Step 4 — Use persistence and minimum-sample rules before escalating; one noisy window is not sufficient evidence

Use persistence and minimum-sample rules before escalating; one noisy window is not sufficient evidence.

Check: Record the observable result or explain the error before continuing.

Step 5 — Open incident_template.md; complete impact, timeline, affected versions, containment, recovery and actions

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Open incident_template.md; complete impact, timeline, affected versions, containment, recovery and actions.

Check: Record the observable result or explain the error before continuing.

Step 6 — Choose containment: rollback candidate, disable corrupt feature or scale capacity, and justify it from the evidence

Choose containment: rollback candidate, disable corrupt feature or scale capacity, and justify it from the evidence.

Check: Record the observable result or explain the error before continuing.

Step 7 — Verify logs and screenshots contain no PII, tokens or full prompts

Verify logs and screenshots contain no PII, tokens or full prompts.

Check: Record the observable result or explain the error before continuing.

Step 8 — Create an executive summary and a technical evidence appendix, then peer-review against the documented acceptance criteria

Create an executive summary and a technical evidence appendix, then peer-review against the documented acceptance criteria.

Check: Record the observable result or explain the error before continuing.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Acceptance checklist

- □ Every required artifact exists and opens without error.
- □ Results are tied to the supplied mock data or an approved training project.
- □ The conclusion cites metrics, IDs or command output rather than opinion alone.
- □ No credentials, tokens, personal data or unredacted sensitive prompts are present.
- □ Cleanup is completed or the retained resource is documented with an owner and expiry.

Troubleshooting

Symptom	Likely cause	Corrective action
403 PERMISSION_DENIED	API disabled, wrong project, or IAM role missing	Verify gcloud config get-value project, API state and the least-privilege role.
429 RESOURCE_EXHAUSTED	Quota or rate limit reached	Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota.
Region/location mismatch	Dataset, bucket, model or endpoint uses another location	Recreate only the training resource in one approved region; do not copy sensitive data casually.
Result differs from expected	Data, dependency or configuration version drifted	Compare file hashes, pinned versions and parameters; retain the diff as evidence.

Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work,

use the Console resource views and activity evidence to verify deletion. Do not delete a shared

project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

Source Register

Authority	Source
content_source	reference/- Master Trainer Slides - Google Professional Machine Learning Engineer Training (Synchronous e-Learning) - v7.pptx
design_source	Current Tertiary all-white house design with editable mechanism diagrams and native charts
exam_guide	https://cloud.google.com/learn/certification/guides/machine-learning-engineer
course_page	https://www.tertiarycourses.com.sg/google-professional-machine-learning-engineer-training.html
pipeline_guide	https://tutorialsdojo.com/a-beginners-guide-to-the-machine-learning-pipeline-on-gcp/
gcp_setup	https://www.kdnuggets.com/5-steps-google-cloud-platform
deployment	https://dvquys.com/posts/deploy-ml-gcp/
gcp_tutorial_pdf	https://deepgraphlearning.github.io/coursewebsite/assets/files/GCP%20tutorial.pdf
deployment_video	https://www.techtarget.com/searchcloudcomputing/video/How-to-deploy-machine-learning-models-on-Googles-AI-Platform
course_reference	https://www.udemy.com/course/machine-learning-on-google-cloud-platform/
vertex_sample	https://github.com/GoogleCloudPlatform/vertex-ai-samples/blob/main/notebooks/official/training/sdk_pytorch_torchrun_custom_container_training_imagenet.ipynb
gke_reference	https://github.com/Shreyjain203/Machine-Learning-Model-to-Production-Website-using-GKE
gpu_reference	https://github.com/JiahongChen/Set-up-deep-learning-frameworks-with-GPU-on-Google-Cloud-Platform
gpu_instance_reference	https://github.com/seem/blog/blob/main/tils/how-to-setup-a-gpu-instance-on-google-cloud-platform.qmd

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

- Open book using approved course slides and Learner Guide; question papers only are learner-facing.

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg